

Trainings ▾

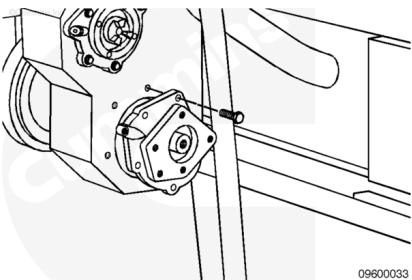
Preparatory Steps

Remove the hydraulic pump. See equipment manufacturer service information.



Remove

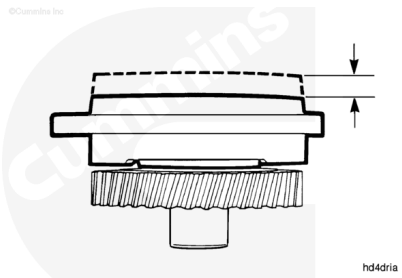
- Remove the hydraulic pump drive mounting capscrews.
- Remove the hydraulic drive assembly.
- Remove and discard the gasket.



Inspect for Reuse

Measure the end clearance.

Hydraulic Pump Drive End Clearance		
mm		in
0.125	MIN	0.005
0.285	MAX	0.011

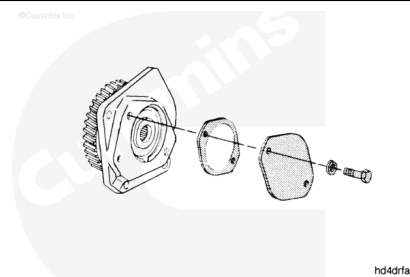


If the end clearance is **not** within specifications, the assembly **must** be rebuilt.

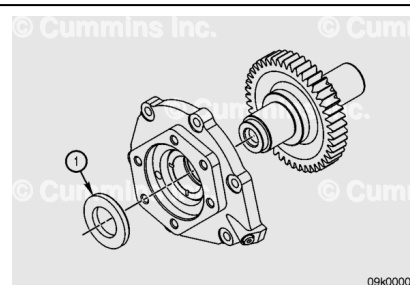
Disassemble

Remove the cover, if equipped.

Remove and discard the gasket.



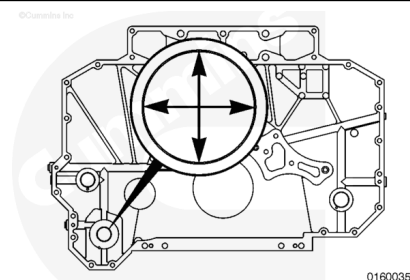
Use a press and suitable mandrel to remove the shaft and gear assembly from the housing. The shaft and gear assembly **must** be pressed out of the retaining ring (1).



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturers recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

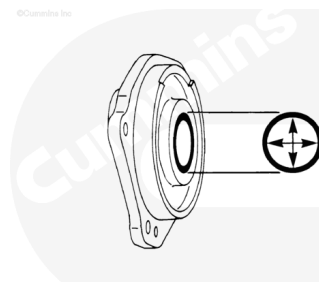
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the hydraulic drive bushing bore in the front gear cover and measure the inside diameter. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>)

Measure the inside diameter of the hydraulic pump drive bushings.

Hydraulic Pump Drive Bushing Inside Diameter		
mm		in
49.95	MIN	1.967
50.03	MAX	1.970

If the inside diameter of either bushing is **not** within specifications, both bushings **must** be replaced.



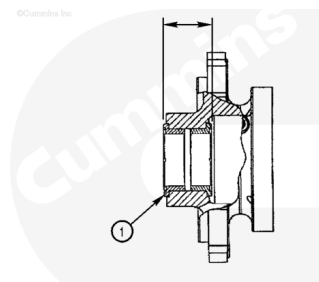
Check the grooved surfaces of the bushings (1) for damage.

Measure the distance from the outside of the thrust faces of the two bushings.

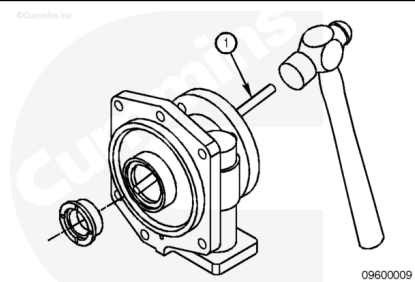
Check the bushings in four locations.

Hydraulic Pump Drive Bushing Thrust Face Distance		
mm		in
45.370	MIN	1.786
45.450	MAX	1.789

If the distance between the two thrust faces is **not** within specification, both bushings **must** be replaced.

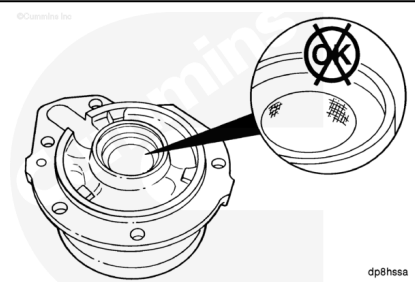


Remove the flanged bushing from the drive housing. Use a brass drift (1), taking care **not** to damage the bushing bore of the housing.



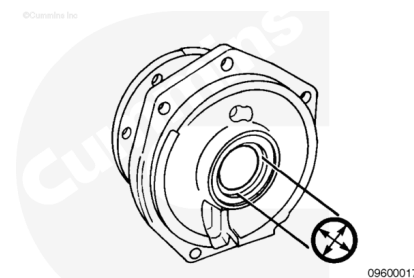
If the bushing was removed, inspect the housing for scoring and other damage.

Replace the housing if damaged.



If the bushing was removed, measure the inside diameter of the bushing bore.

Hydraulic Pump Drive Unit Housing Bore Inside Diameter		
mm		in
60.00	MIN	2.362
60.05	MAX	2.364

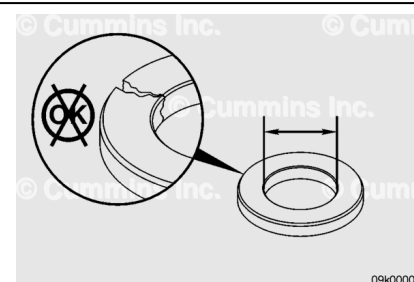


Replace the housing if the bushing bore is **not** within specifications.

Inspect the retaining ring for any signs of damage or cracks. Replace, if necessary.

Measure the inside diameter of the retaining ring.

Hydraulic Pump Drive Retaining Ring Inside Diameter		
mm		in
41.869	MIN	1.6484
41.893	MAX	1.6493

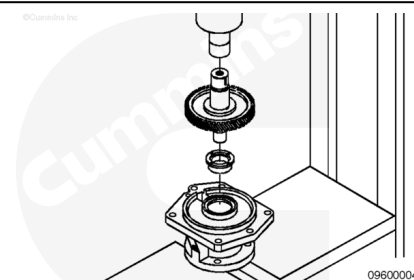


Replace the retaining ring if the internal diameter is **not** within specification.

Assemble

⚠ CAUTION ⚠

The bushing must be pressed in square to the bores of the housing. If the bushing is not square, damage to the bushing could possibly occur.



Install the bushing into the housing.

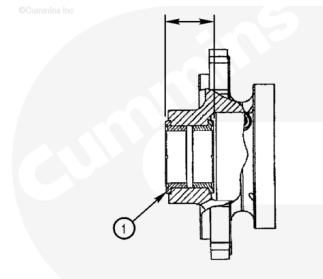
Support the housing on a press. Use the shaft as a mandrel to press the bushing into position.

Turn the housing over. Press the remaining bushing into place.

Measure the distance from the outside of the thrust faces of the two bushings. Check in four positions.

Hydraulic Pump Drive Bushing Thrust Face Distance

mm		in
45.370	MIN	1.786
45.450	MAX	1.789



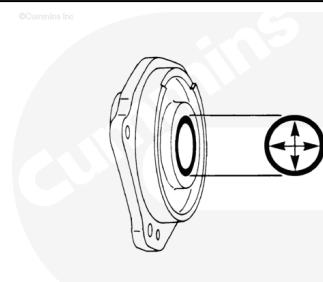
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If the distance between the two thrust faces is **not** within specification, check the installation. Make sure the bushings are correctly positioned.

Lubricate the bushings. Insert the drive shaft assembly into the housing to verify it spins freely.

Measure the inside diameter of the hydraulic pump drive unit bushings to check for any distortion during installation.

Hydraulic Pump Drive Bushing Inside Diameter		
mm		in
49.95	MIN	1.967
50.03	MAX	1.970



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If the inside diameter of either bushing is **not** within specifications, both bushings **must** be replaced.

Note : The hydraulic pump drive unit housing must be inspected again to identify any reason for distortion.

Lubricate the bushing. Insert the drive shaft assembly into the housing to verify it spins freely.

Lubricate the grooved surface of the thrust bearing with Lubriplate™ 105, Part Number 3163086, or equivalent.

Install the gear and shaft assembly in the housing.

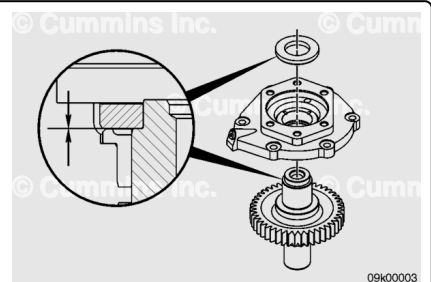
Move the assembly to a mandrel press.

Position the assembly with the gear on the underside of the housing.

Make sure the assembly is supported squarely on the end of the shaft.

Position the retaining ring on the shaft with the beveled edge facing down toward the bushing.

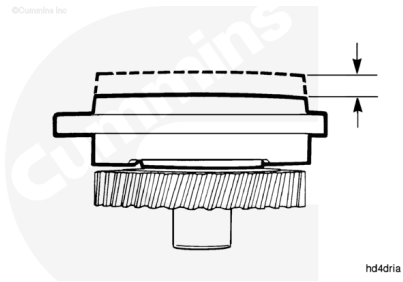
Use the press and a suitable mandrel to press the retaining ring onto the shaft until it is against the shoulder on the shaft.



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Measure the end clearance.

Hydraulic Pump Drive End Clearance		
mm		in
0.125	MIN	0.005
0.285	MAX	0.011



If the end clearance is **not** within specifications, the assembly **must** be inspected to check for correct installation of the bushings and retaining ring.

Note : Spin the shaft assembly in the housing to verify that it spins freely.

Install

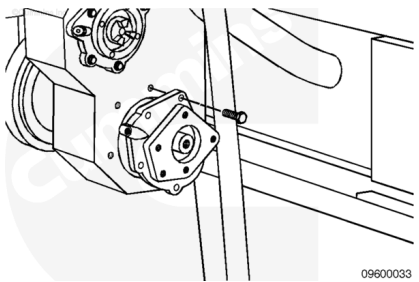
Lubricate hydraulic drive bushing in the front cover with Lubriplate™ 105.

Install the new gasket.

Install the pump drive.

Align the capscrew holes.

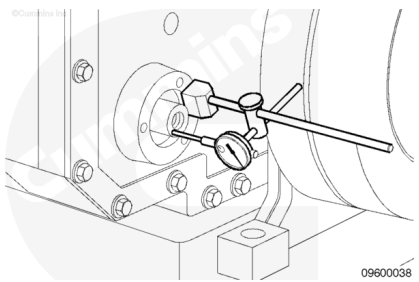
Install and tighten the hydraulic pump drive mounting capscrews.



Torque Value: 45 n•m [33 ft-lb]

Measure the hydraulic pump drive end clearance after installing the drive.

Hydraulic Pump Drive End Clearance - Installed		
mm		in
0.125	MIN	0.005
0.285	MAX	0.011

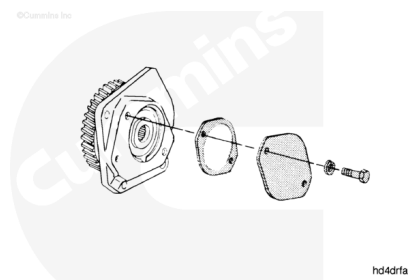


If the end clearance is **not** within specifications, the assembly **must** be inspected for correct installation.

Install a new gasket. Install the cover, if equipped.

Install the locktight capscrews or, if fitted, lock washers and the capscrews.

Torque Value: 27 n•m [239 in-lb]



Finishing Steps

Install the hydraulic pump. See equipment manufacturer service information.

